Enjambre ABCDS Project

Author: Santiago González · Oct 18, 2026

Public record: DOI 10.5281/zenodo.23448232 (Zenodo, CC BY-NC-ND 4.0 license). English translation of the original Spanish document.

Summary

Enjambre ABCDS is a space data center made of a huge number of simple, heat-tolerant chips, spread over a very large area and organized like a company. This document records the design as proposed by its author, Santiago González, on October 18, 2026.

The core idea is to do the opposite of the industry: instead of packing very powerful chips into a small space and fighting to cool them, use weaker chips that tolerate heat, spaced apart, so each one cools itself by radiating into space.

The system has five layers, named after the positions in a company:

- **A, employees:** the chips that do the work.
- **B, line supervisors:** they package what their A chips produce and pass it up, nothing more.
- **C, board of directors:** several directors specialized by function that receive, verify and sign the results.
- **D, bylaws:** the internal rulebook of each C, isolated and almost never updated.
- **S, stores:** the entry and exit point for customers, unable to alter what it delivers.

The problem

Data centers spend a huge share of their energy and money removing heat. The problem is not the total amount of heat but its density: a lot of heat packed into very little space.

Space looks like the answer because it is cold, but not directly:

- In a vacuum there is no air or water to carry heat away. It can only be shed by radiation, which is slow.
- One square meter of radiator at room temperature sheds about 400 to 450 W. A 1 MW data center would need about 2,000 to 2,500 m² of radiators.
- The Sun heats everything it touches, so shade is needed.

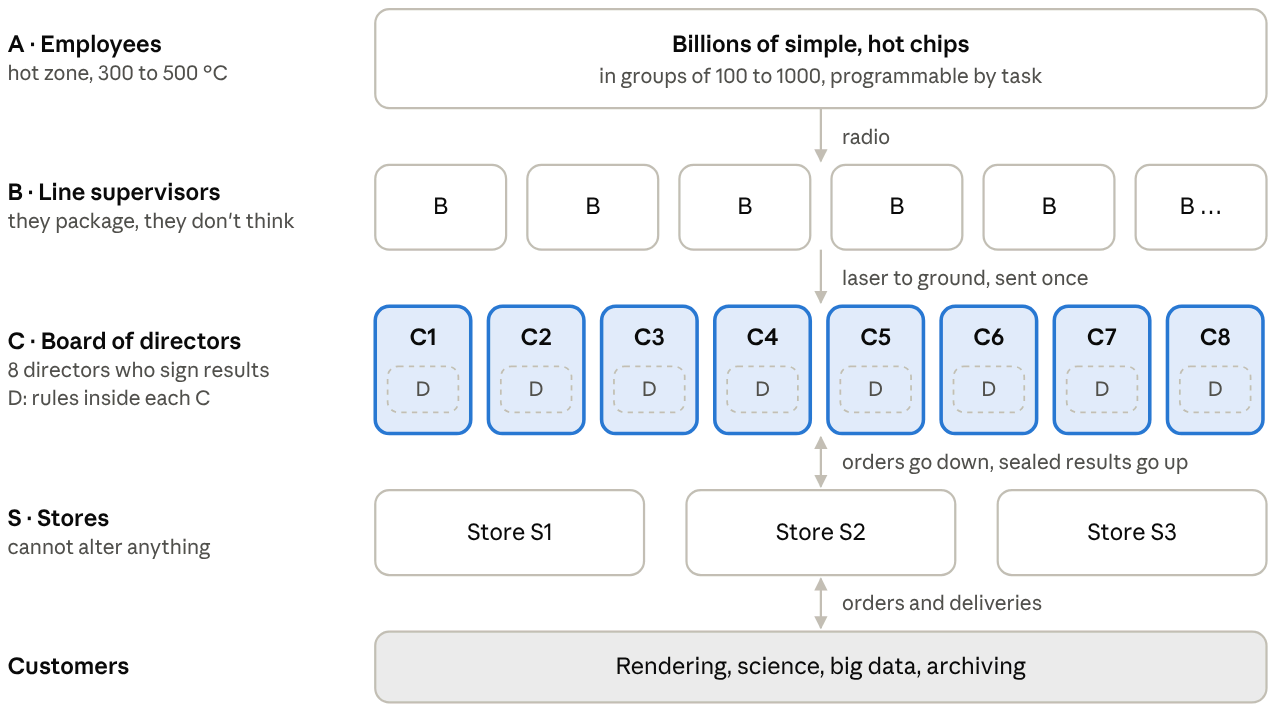
The cold places of the solar system (Uranus, Triton) don't work: Uranus has no surface and its interior is about 5,000 °C, Triton has almost no atmosphere, and both are about 4 light-hours from Earth.

Design principles

1. **Spread out instead of concentrate.** Space is plentiful. If the chips are spaced apart, each one has its own patch of cold sky to radiate into, and heat density drops.
2. **Weak but tough chips.** Less powerful chips are accepted in exchange for tolerating 300 to 500 °C. Radiation grows with the fourth power of temperature: a radiator at 500 °C sheds about 40 times more heat per square meter than one at room temperature.
3. **Zones at different temperatures.** Only the A chips run hot. B, C and S live in the shade or on the ground, with normal electronics.
4. **Compartmentalize.** Each part controls only its own area. A failure or an attack stays where it started.
5. **If it works, don't touch it.** The rulebook (D) is updated only in emergencies, manually, from the ground.

Layered architecture

The system works like a company: the factory produces, the supervisors report, the board verifies and the stores sell. No layer skips another.



ABCDS architecture · 5 layers

The dashed boxes inside each C are its D: the internal rulebook, with no connection to the others.

Layer	Role in the company	What it does	Where it lives
A	Employees	Compute. Each group of 100 to 1000 chips has a fixed, programmable composition (for example, 100 on one task, 100 on another)	Orbit, hot zone
B	Line supervisors	Bundle their A chips' output into a box with a fixed schema, mark the gaps left by dead chips and send it. Nothing more	Orbit, in the shade

Layer	Role in the company	What it does	Where it lives
C	Board of directors	8 directors specialized by function (storage, compute, monitoring, control). All receive everything, each controls its own B nodes and signs its results	Earth and Moon, at separate sites
D	Bylaws	Identical rulebook inside each C, with no network between them. Updated only in emergencies, manually	Inside each C
S	Stores	Take orders and deliver sealed results. They can package but not modify	Earth, several

Why no labels are needed: since each group of A chips always has the same composition, a piece of data's position in the box already tells you what it is, just like a column in a spreadsheet.

Physical infrastructure

A single part, the mirror, provides both shade and power. The rest of the infrastructure moves heat and data.

Part	Function	Detail
Mirror	Reflects sunlight instead of absorbing it and redirects it to the panels	Shades B and the sensitive electronics. Must point accurately and degrades over time
Solar panels	Turn the concentrated light into electricity	Convert 30 to 40% of the light; the rest is heat, so they carry their own radiators
Radiators	Shed the chips' heat into space	Smaller because A runs hot: at 500 °C each m² sheds about 40 times more than at room temperature

Part	Function	Detail
Radio A → B	Communication within each group	The A chips take turns to transmit, or each uses its own frequency. Within one chip or board, short ceramic wires are used
Laser B → Earth	Downlink of the boxes	About 100 Gbps per link. Clouds block it, so there are several stations at dry sites
Fiber on the ground	Copies each box to the 8 C nodes	Each box comes down from space only once; copying on the ground is cheap

Request flow

An order travels from the store down to the chips, and the result comes back sealed along the same path.

1. The customer places an order at any store S.
2. S passes it to the control director among the C nodes.
3. Following their D, the C nodes decide which groups of A will do it and send the order up.
4. The A chips compute. Each B bundles its A chips' results into a box with a fixed schema and marks the gaps.
5. B waits its turn and sends the box by laser to whichever ground station has clear skies.
6. On the ground, the box is copied by fiber to all 8 C nodes.
7. Each C reviews it according to its function, and the C nodes agree on what arrived and in what order.
8. The C nodes sign the result and hand it to S.
9. S packages and delivers it. The customer checks the signatures: if anything was altered along the way, they don't match.

The intermediate layers can summarize instead of forwarding everything. That way the result comes down to Earth, not the drafts.

Security and fault tolerance

The system keeps delivering correct results even if 2 of the 8 directors are hacked or any store goes down.

Threat	Defense
A chips die	There are millions more. B marks the gap in the box so nobody mistakes empty for error
A cloud blocks a laser	Several ground stations at dry, distant sites; they are never all covered at once
A C node loses a box	The C nodes agree on what arrived and in what order before acting
A C node is hacked	Its D is isolated: the attacker cannot rewrite the others' rulebook
A hacked C lies	With 8 directors, agreement among the honest ones withstands up to 2 traitors (at least 3 × traitors + 1 nodes are needed)
Someone tampers with a D	Each C publishes the fingerprint (hash) of its D; a different fingerprint reveals the change without connecting the D's
A store S is hacked	S cannot modify results: they are signed by the C nodes and the customer checks the signature. There are several S

Updates: D is almost never touched. In an emergency it is updated manually from the ground, one at a time. It is slow on purpose; the cost is that security holes found later take longer to close.

Approximate numbers

Running hot shrinks the radiators about 44 times; the real bottleneck is getting the data down to Earth. These are napkin calculations, not measurements.

Calculation	Result	Assumption
Heat radiated by 1 m² at 27 °C	~460 W	Stefan-Boltzmann law, emissivity 1, one side
Heat radiated by 1 m² at 500 °C	~20,000 W (about 44 times more)	Same as above
Radiators for 1 MW	~2,200 m² at 27 °C; ~50 m² at 500 °C	Not counting heat arriving from the Sun
Water to cool 1 MW by evaporation	~38 tonnes per day	Heat of vaporization of water, 2.26 MJ/kg
Traffic from a trillion chips	~1 petabit per second	10^{12} chips sending 1 kbps each
Laser links for that traffic	~10,000	100 Gbps per link, no summarizing
Hierarchy levels for a trillion chips	4	Groups of 1000 at each level
Mass of 1 MW in orbit	10 to 20 tonnes	10 to 20 kg per kW including panels and radiators
Cost to launch 1 MW today	US$15 to 60 million	US$1,500 to 3,000 per kg
Hacked directors tolerated	2 of 8	Byzantine agreement: needs 3 × traitors + 1

As a published reference: an analysis of SpaceX's plan estimates about 41 m² of radiator for a 100 kW satellite radiating at 400 K, versus about 834,000 m² for a single 1 GW facility (source). Google estimates launch could cost under US$200 per kg by the mid-2030s (source), which would bring the cost of launching 1 MW down to about US$2 to 4 million.

Open problems

The biggest problem is the A chip: the ones that tolerate 500 °C today are millions of times slower than a GPU.

1. **Slow A chips.** NASA's silicon carbide circuits ran for almost 22 days at about 460 °C in Venus-like conditions, but they were test circuits with dozens of transistors (source). A modern GPU has tens of billions.
2. **Hot memory.** No high-capacity memory works at 500 °C. Flash memory fails above about 150 °C.
3. **Solder, insulation and wiring.** Standard solder melts at 180 to 220 °C; copper at 500 °C has about 3 times more resistance. Ceramic materials are needed.
4. **Launch mass and cost.** With slow chips, many more kilograms are needed per unit of compute.
5. **Space debris.** Millions of satellites raise the risk of cascading collisions and need permits, control and end-of-life disposal.
6. **Mirror.** It must point accurately at all times and degrades from radiation and micrometeoroids.
7. **Type of workload.** It suits batch jobs that split into independent parts. It does not suit workloads where chips must talk to each other constantly, such as training large AI models.

Unanswered questions:

- Which concrete workload justifies using slow chips that are cheap to cool?
- Where exactly does B go: on the same satellite as its A chips, or on a separate one in the mirror's shade?
- What functions do the other 4 directors have besides storage, compute, monitoring and control?

Improved version: the 125 °C middle ground

The best answer to the chip problem is to avoid the extreme: run A at about 125 °C instead of 500 °C. That allows hardened, nearly modern silicon chips, and the radiators are still about 3 times smaller than at room temperature.

Option for A	Radiators for 1 MW	Available chips	Available memory
27 °C, as today	~2,200 m²	Modern GPUs and processors	All standard memory

Option for A	Radiators for 1 MW	Available chips	Available memory
125 °C, middle ground	**~700 m²**	**Automotive and industrial grade silicon, already made for 125 to 150 °C**	**Automotive grade memory up to about 105 to 125 °C**
500 °C, extreme	~50 m²	Silicon carbide, circuits with dozens of transistors	No high-capacity memory exists

Why the middle ground:

- **It solves the biggest open problem** (chips millions of times slower) in exchange for larger radiators.
- **It can be built with parts that already exist,** without waiting for 500 °C electronics to mature.
- **It keeps the full ABCDS architecture:** only A's operating temperature changes.

Efficiency: per watt, these chips still do less work than the best ones in a ground data center. The system makes up for it with free solar power, cooling with no water and no extra energy use, and no need for land or a power grid. What decides whether it pays off is compute per kilogram launched, which improves as launch prices fall.

Prior work and what this design adds

Spreading a data center across many satellites is already being pursued by SpaceX, Google, Starcloud and others, but all of them use powerful chips at normal temperatures. I found no one proposing, on purpose, weak and hot chips spread out with this hierarchy.

Work	Year	Similarity	Difference
LEACH (Heinzelman, Chandrakasan, Balakrishnan)	2000	Grouped sensors with a cluster head that gathers, compresses and forwards, like B	Ground sensors, no board of directors, isolated rulebook or stores
Matrioshka brain (Robert Bradbury)	1997	Massive computing in space that reuses heat in layers	A sphere around an entire star; theoretical concept with no network architecture

Work	Year	Similarity	Difference
Project Suncatcher (Google)	2025	Many satellites working as a single center, linked by laser	Uses standard TPUs in tight formation (81 satellites 100–200 m apart); prototypes planned for 2027
Starcloud	2025–2026	Satellites with GPUs cooled only by radiation	Nvidia GPUs; already launched a satellite with an H100 and plans 88,000
SpaceX FCC filing	2026	Spreading heat across many satellites so each stays manageable	Up to one million satellites of about 100 kW connected to Starlink
Ramon.Space DCiS	2026	Fault-tolerant micro data center spread across small satellites	Standard modules and virtualization; no layered hierarchy described
Orbital	2026	Many independent nodes that scale by adding satellites	Over 100,000 satellites of 100 kW for AI
NASA Glenn SiC chips	2016	Electronics that run at 460–500 °C	Built for Venus probes, not data centers

What this design combines that does not appear in those works:

- **Weak, hot chips on purpose,** spaced apart so heat density drops, instead of GPUs that must be cooled.
- **Zones at different temperatures:** only A runs hot; B, C and S use normal electronics in the shade or on the ground.
- **The full company hierarchy:** B with no intelligence and a fixed schema, a board of 8 directors by function, an isolated rulebook D in each one, and stores S that cannot alter signed results.

Not finding it in a search does not prove nobody has thought of it. Publishing this document with a date serves as a public record of the idea.

Sources consulted on October 18, 2026: those linked in the table and in the numbers section.